Software Test Plan (STP) – Online Quiz System

Project: Online Quiz System
Version: 1.0
Authors: PAVAN KUMAR K M (PES2UG24CS646), ROSALIN VERMA (PES2UG24CS655), S SHREYAS KUMAR (PES2UG24CS656), VAGEESH (PES2UG24CS663)

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1. Introduction

Purpose: This document defines the test plan for the **Online Quiz System v1.0**. It outlines the testing objectives, scope, test strategy, resources, schedule, responsibilities, and acceptance criteria for verifying the system.

The Online Quiz System is a console-based C++ client-server application that allows students to register and log in, access online MCQ quizzes, answer and submit quizzes, receive scores, view previous results, and view leaderboards. Administrators can manage quiz questions and view student results. The system communicates with an online server and data store.

Scope: Testing covers the major Online Quiz System features, including:

- Student registration and login
- Quiz selection and retrieval
- MCQ display and answer validation
- Quiz submission
- Score calculation and result display
- Online result storage
- Previous result viewing
- Leaderboard display
- Administrator question and result management
- Performance, usability, reliability, scalability, auditability, and security requirements

The system is required to operate as a C++ console application with Internet connectivity.

References:

- Online Quiz System SRS v1.0
- Online Quiz System Design Specifications v1.0
- Requirements Traceability Matrix (RTM)
- Test Cases and Test Data

Definitions:

- **OQS** – Online Quiz System
- **MCQ** – Multiple Choice Question
- **SRS** – Software Requirements Specification
- **FR** – Functional Requirement
- **NFR** – Non-Functional Requirement
- **RTM** – Requirements Traceability Matrix
- **UAT** – User Acceptance Testing
- **API** – Application Programming Interface
- **HTTPS** – Hypertext Transfer Protocol Secure

2. Test Items

- Student Registration Module
- Student Authentication/Login Module
- Quiz Selection Module
- Quiz Retrieval Module
- MCQ Display Module
- Answer Validation Module
- Quiz Navigation Module
- Quiz Submission Module
- Score Calculation Module
- Result Display Module
- Result Storage Module
- Previous Results Module
- Leaderboard Module
- Administrator Question Management Module
- Administrator Result Management Module
- Online Server/API communication
- Security and audit logging

3. Features to be Tested

Features mapped to SRS requirement IDs:

Functional Requirements

- **OQS-F-001:** Student registration with unique username and password
- **OQS-F-002:** Student authentication/login
- **OQS-F-003:** Display available quizzes
- **OQS-F-004:** Select an available quiz
- **OQS-F-005:** Retrieve quiz questions and options
- **OQS-F-006:** Display one MCQ at a time
- **OQS-F-007:** Validate student answers
- **OQS-F-008:** Navigate through all quiz questions
- **OQS-F-009:** Submit completed quiz
- **OQS-F-010:** Calculate quiz score
- **OQS-F-011:** Display final score and result summary
- **OQS-F-012:** Store quiz result on the online server
- **OQS-F-013:** View previous quiz results
- **OQS-F-014:** Display quiz leaderboard
- **OQS-F-015:** Administrator question and result management

Non-Functional Requirements

- **OQS-NF-001:** Normal quiz and result requests should receive a server response within 3 seconds.
- **OQS-NF-002:** Console interface shall provide clear menus, prompts, and error messages.
- **OQS-NF-003:** Online service shall target at least 99% monthly availability excluding planned maintenance.
- **OQS-NF-004:** System shall support at least 50 concurrent student sessions.
- **OQS-NF-005:** Important events such as login attempts, quiz submissions, and administrator changes shall be timestamped in logs.

Security Requirements

- **OQS-SR-001:** HTTPS/TLS shall be used for client-server communication.
- **OQS-SR-002:** Passwords shall be securely hashed and shall not be stored in plaintext.
- **OQS-SR-003:** Administrator functions shall be restricted to authenticated administrator accounts.
- **OQS-SR-004:** Quiz submissions shall be validated against the authenticated student.
- **OQS-SR-005:** Failed logins and administrator actions shall be logged without recording passwords.

4. Features Not to be Tested

- Internal implementation details of the online server that are not exposed through the defined API.
- Third-party infrastructure or services outside the control of the Online Quiz System.
- Physical hardware other than the standard computer, keyboard, display, and network connection required to operate the console application.
- Operating-system internals.
- Internet Service Provider infrastructure.
- Database internals beyond verifying that required quiz and result data are correctly stored and retrieved.

5. Test Approach / Strategy

1.Testing Levels

Unit Testing

Individual modules will be tested independently.

Examples:

- Registration validation
- Login authentication
- Answer validation
- Score calculation
- Input validation

Integration Testing

Interactions between system components will be tested.

Examples:

- C++ client ↔ Online Quiz Server/API
- Login module ↔ User data store
- Quiz module ↔ Quiz data store
- Submission module ↔ Result storage
- Administrator module ↔ Question/result data

System Testing

The complete Online Quiz System will be tested as an integrated application.

Examples:

- Student registration → login → quiz selection → quiz attempt → submission → score → result storage
- Administrator login → question management → result viewing

Acceptance Testing

The system will be tested against the acceptance criteria specified in the SRS to determine whether it is ready for use.

2.Types of Testing

Functional Testing

Functional testing will verify that all OQS functional requirements behave as specified.

Regression Testing

Previously tested functionality will be retested after changes or defect fixes to ensure that existing features continue to work.

Performance Testing

Performance testing will verify:

- Server response time
- Quiz retrieval performance
- Result submission performance
- Support for at least 50 concurrent student sessions

Usability Testing

The console interface will be checked for:

- Clear menus
- Understandable prompts
- Clear MCQ options
- Appropriate error messages
- Easy navigation

Security Testing

Security testing will verify authentication, password protection, authorization, secure communication, submission validation, and audit logging.

Reliability Testing

The online service will be tested against the availability requirement of at least 99% monthly availability, excluding planned maintenance.

3.Entry Criteria

Testing shall begin when:

- A stable build of the Online Quiz System is available.
- Required modules are implemented.
- Test data is available.
- The online server/API is available.
- The test environment is configured.
- Required database/data-store access is available.
- Test cases have been reviewed and approved.

4.Exit Criteria

Testing shall be completed when:

- 100% of planned test cases have been executed.
- All high-priority functional requirements have been tested.
- No critical defects remain open.
- Required acceptance criteria are satisfied.
- Major security requirements have been verified.
- Performance requirements have been tested.
- Test results and defect reports have been documented.

5.1 Security Validation

- Verification of HTTPS/TLS communication.
- Verification that passwords are not stored in plaintext.
- Testing of secure password hashing.
- Testing of administrator authorization.
- Testing that students cannot access administrator functions.
- Verification that quiz submissions belong to the authenticated student.
- Testing of failed-login logging.
- Testing of administrator-action logging.
- Verification that passwords are not written to audit logs.

- Input validation for registration, login, quiz answers, and administrative operations.

6. Test Environment

Hardware:

- Computer/laptop
- Keyboard
- Display
- Network connectivity

Software:

- C++ Online Quiz Client v1.0
- Online Quiz Server/API
- Online database/data store
- Windows or Linux operating system
- Internet connection

Tools:

- C++ unit testing framework for unit tests
- Postman for API testing
- JMeter or similar tools for performance/load testing
- Git for version control
- Jira or similar issue-tracking tool for defect management

Test Data:

- Valid student usernames and passwords
- Duplicate usernames
- Invalid login credentials
- Sample quizzes
- MCQ questions with multiple options
- Correct and incorrect answers
- Completed and incomplete quizzes
- Student result records
- Administrator accounts
- Invalid user inputs
- Concurrent student sessions

7. Test Schedule

Milestones:

- Test case design: 28-Sep-2026
- Environment setup: 30-Sep-2026
- Test execution start: 1-Oct-2026
- Test execution end: 13-Oct-2026
- UAT: 15-Oct-2026 to 22-Oct-2026

8. Test Deliverables

- Software Test Plan
- Test Cases
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix
- Performance Test Results
- Security Test Results
- Test Summary Report
- Acceptance Test Results

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Pavan Kumar K M	Prepare plan, coordinate execution
Test Engineer	Vageesh	Design & execute test cases, log defects
Developer	Rosalin Verma	Support defect fixes and triage
Product Owner	S Shreyas Kumar	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early builds and perform smoke testing
Online server/API unavailable	Maintain a backup test environment or mock API
Internet connectivity issues	Maintain an alternative network connection for testing
Insufficient test data	Prepare test data before test execution
Database/data-store failure	Maintain backup test data and database recovery procedures
High number of defects	Prioritize critical defects and perform regression testing
Performance degradation with multiple users	Conduct load testing and optimize server/API performance
Unauthorized access to admin functions	Perform authorization and security testing

11. Assumptions & Dependencies

- The online server/API will be available during testing.
- Required test accounts will be provided before testing.
- Test quiz data will be available.
- The online database/data store will be accessible.
- Internet connectivity will be available.
- The C++ client will be compatible with the intended Windows/Linux environment.
- Required APIs and networking components will be available.
- Administrator credentials will be available for authorized testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The online server/API is unavailable for an extended period.
- The test environment becomes unavailable.
- A build contains critical defects that block more than 30% of planned test cases.
- The database/data store is unavailable.
- Critical authentication or communication failures prevent further testing.

Resume testing if:

- Blocking defects have been resolved.
- The online server/API becomes available.
- The test environment is restored.
- Required test data is available.
- A stable build is provided.

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.
Example:

- **OQS-F-001 (Register user)** → TC-REG-01, TC-REG-02
- **OQS-F-002 (Login)** → TC-AUTH-01, TC-AUTH-02
- **OQS-F-005 (Fetch MCQs)** → TC-QUIZ-01, TC-QUIZ-02
- **OQS-F-007 (Validate answer)** → TC-ANS-01, TC-ANS-02
- **OQS-F-009 (Submit quiz)** → TC-SUB-01, TC-SUB-02
- **OQS-F-010 (Calculate score)** → TC-SCORE-01
- **OQS-F-012 (Store result)** → TC-RESULT-01
- **OQS-F-015 (Admin functions)** → TC-ADMIN-01, TC-ADMIN-02
- **OQS-NF-001 (Response time)** → TC-PERF-01
- **OQS-NF-004 (50 concurrent sessions)** → TC-PERF-02
- **OQS-SR-001 (HTTPS/TLS)** → TC-SEC-01
- **OQS-SR-002 (Password hashing)** → TC-SEC-02

The SRS RTM already maps major requirements to modules and test cases, including registration, login, quiz, submission, scoring, results, administrator functions, performance, and HTTPS/TLS.

14. Test Metrics & Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage of test cases passed
- Percentage of test cases failed
- Percentage of test cases blocked
- Defect count
- Defect density
- Defect severity distribution
- Defect aging
- Requirement coverage
- Functional requirement coverage
- Security requirement coverage
- Average server response time
- Percentage of requests meeting the 3-second response-time requirement
- Concurrent-user performance
- Test execution progress

Reports:

- Daily Test Execution Report
- Defect Report
- Performance Test Report
- Security Test Report
- Requirements Coverage Report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Pavan Kumar K M	Pavan / 16/9/2026
Test Engineer	Vageesh	Vageesh/ 16/9/2026
Dev Lead	Rosalin Verma	Rosalin/ 16/9/2026
Product Owner	S Shreyas Kumar	Shreyas/ 16/9/2026